

Cross-national comparative assessment of synergies and conflicts in climate policy mixes

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Carbon pricing is one of the most effective climate mitigation instruments, yet how it interacts with other climate policies and shapes their effectiveness remains unclear. Integrating the most comprehensive climate policy datasets covering more than 10,000 climate policies over 100 countries, this paper quantifies such policy interaction effects. We use synthetic control methods for policy effect estimation, construct a Global Climate Policy Index to reflect different policy designs and conduct global comparative studies on policy interactions. Our results show that under historical average levels of prices and coverage, emissions trading systems and carbon taxes reduce emission intensity by approximately 15.4% and 8.5%, respectively, and these reductions are significantly influenced by policy interactions, with both synergies and conflicts based on market maturity and policy intensity. Counterfactual simulations suggest that reducing such policy conflicts could improve carbon pricing effectiveness by up to 22.3%, highlighting the importance of policy coherence in climate mitigation strategies.

Global carbon emissions have accelerated since the 1950s and have kept rising in the past seven decades¹. This continuous upward trend has intensified climate change risks to economies, societies and ecosystems, highlighting urgent international efforts to mitigate global warming^{2–5}. Assessment studies have shown that the Nationally Determined Contributions under the Paris Agreement are insufficient to limit global surface temperature increases to well below 2 °C (ref. 6), indicating the requirement of stronger and more effective climate policies⁷. Among the various available tools, carbon pricing is widely recognized as the most cost-efficient instrument for accelerating the low-carbon transition^{8,9}.

Currently, 51 countries—38 developed and 13 developing—have implemented over 80 carbon pricing policies¹⁰, covering more than 28% of global emissions. However, the practical effectiveness of these

policies is highly contingent on the policy element design, the institutional settings and the implementing contexts^{11,12}. For example, differences in allowance allocation, market maturity and complementary measures help explain the heterogeneity in designs and outcomes of the European Union (EU) Emissions Trading System (ETS) and China's national carbon market^{13,14}. Moreover, carbon pricing policies are rarely implemented in isolation but typically coexist with regulatory mandates, subsidies or government procurement programmes¹⁵. Such coexistence may either reinforce or undermine pricing effectiveness^{16–18}. Evidence from the EU ETS suggests that renewable energy subsidies can weaken the effectiveness by distorting market signals¹⁹, whereas carbon taxes combined with green procurements can create reinforcing incentives for low-carbon investment and behaviour^{18,20–26}. These interaction mechanisms imply that policy coherence, rather than

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instrument stringency alone, is central to explaining cross-country differences in carbon pricing outcomes.

Despite the increasing interest in the interactions of carbon pricing and other climate policies²², systematic global evidence on interaction mechanisms remains limited. First, most empirical studies are either case-based or sector-specific^{8,23,27–29}, and conventional econometric approaches such as panel regression and difference-in-differences (DID) provide limited comparative insight into interaction patterns across diverse climate policy portfolios. The lack of external validity makes it difficult and often unreliable to extrapolate these findings to other countries or sectors. Second, climate policy contexts are heterogeneous³⁰ and thus not directly comparable^{31–35}. Simulation-based approaches such as computable general equilibrium models offer alternative frameworks but struggle with data-intensive calibration requirements and are weak in capturing policy heterogeneities and interacting features across countries^{36–38}. A comprehensive and standardized policy database is therefore needed to support robust global comparative analysis of climate policy interactions.

Understanding the synergistic and conflicting effects within climate policy mixes is crucial not only for improving mitigation efficiency but also for expanding carbon pricing adoption. So far, most developing countries still primarily rely on non-carbon-pricing instruments¹⁵ due to political barriers, concerns about potential losses in economic competitiveness, risks of carbon leakage, equity considerations and other issues^{36,39,40}. For these countries, insights derived from international experiences of policy interactions are highly valuable for designing effective and politically feasible carbon pricing policies⁴¹.

A robust assessment of climate policy effectiveness should integrate diverse mitigation strategies implemented worldwide. While quasi-experimental methods have been applied to evaluate individual policies⁸, global evidence on the interactions among multiple instruments remains scarce^{22,42}. Addressing this gap requires a unified framework enabling systematic comparison of policy combinations across countries. In this paper, we develop such a global comparative framework and analyse how different policy mixes affect the performance of carbon pricing policies. This study focuses on carbon pricing policies—taxation and ETSs—given their demonstrated cost-effectiveness and central role in contemporary climate policy frameworks, and quantitatively analyses their interactions with complementary mitigation instruments. To provide systematic global comparisons, this study integrates two carbon pricing policy databases with a global climate policy database^{10,43}, which incorporates over 10,000 climate policies from more than 100 countries over the past two decades⁴⁴. Using a synthetic control analytical framework, we first assess the emission reductions from different carbon pricing implementations country by country, which allows policy-linked outcomes to be obtained at a country-year level. To ensure comparability across heterogeneous policy contexts, we then construct a Global Climate Policy Index, quantifying policy intensity, coverage and enforceability^{45,46}. Finally, we employ these indices to evaluate how different policy instruments interact with carbon pricing to shape its mitigation effects, revealing systematic patterns of synergy and conflict across policy frameworks (Extended Data Fig. 1). Moreover, this study analyses how policy interactions are linked to the observed effectiveness of these policies and demonstrates how climate policy outcomes vary across different policy frameworks (Supplementary Section A).

This study underscores the importance of policy interactions and offers a cross-country framework for climate policy design with three main contributions: (1) providing global evidence on how policy interactions are associated with carbon pricing outcomes, (2) proposing a unified quantitative framework to compare diverse policy regimes and (3) delivering actionable insights for optimizing policy mixes to enhance the adoption and impact of carbon pricing policies globally. These contributions build an empirical foundation for designing synergistic strategies that accelerate global decarbonization.

Global comparative analysis of emission reduction effects

Emission reductions attributable to carbon pricing for each country are assessed using a synthetic control method, which constructs a counterfactual baseline by matching pretreatment emission trends and national characteristics for comparison (Fig. 1). Post-adoption emission intensity drops approximately 15.44% on average across countries implementing ETSs and 8.52% for countries adopting carbon taxes. In the absence of carbon pricing, emission intensity tends to decline only gradually, primarily driven by autonomous technological progress accompanying economic development. The flatness of the synthetic trajectories reflects gradual natural decarbonization, supported by strong pretreatment alignment and extensive robustness and placebo analyses. Complementary analyses extended to other mitigation instruments—including subsidies, regulations, information programmes and government provision policies—provide additional context for their emission impacts (Supplementary Section B).

However, cross-country comparison of policy interaction patterns remains challenging, as countries differ in the intensity, coverage and enforceability of their policies. Simply comparing nominal policy stocks or carbon price levels may fail to capture the broader policy context in which carbon pricing operates. To enable systematic cross-country comparison, we constructed a Global Climate Policy Index, which captures the broader policy context and allows carbon pricing to be evaluated alongside complementary instruments under diverse national conditions. At the global level, ETSs expanded mainly after 2005, whereas carbon taxes were introduced earlier by several European countries beginning in the early 1990s (Fig. 2). Global coverage of carbon pricing increased steadily over time, with a temporary slowdown around 2010. However, carbon tax prices have risen steadily during this period, while ETS prices have fluctuated considerably—reflecting fundamental differences between these two carbon pricing policies.

Cross-country heterogeneity in policy mixes emerges when these indices are mapped globally (Extended Data Fig. 2). Medium- to high-intensity carbon pricing policies are concentrated in Europe, with Norway, Sweden and Finland leading. Outside Europe, the USA and Canada also display relatively well-developed mitigation policy portfolios. In contrast, most African countries exhibit lower overall policy intensity. Across regions, government provision and regulatory instruments constitute the largest shares of national policy portfolios, whereas information policies are less prevalent and subsidy policies—capturing support for renewable energy production and technology adoption—remain comparatively limited (Supplementary Tables 18 and 19). These cross-sectional patterns motivate the subsequent analysis of how different policy instruments are associated with synergistic or conflicting relationships with carbon pricing.

Synergistic and conflicting interactions with other policies

The emission-intensity changes estimated for carbon pricing using the synthetic control approach reflect not only the effects of pricing instruments themselves but also their interactions with pre-existing climate policies. We therefore took these estimated reductions as the outcome variable in a cross-national regression to examine how additional policy instruments are linked to the observed effectiveness of carbon pricing (Extended Data Table 1). The positive or negative regression coefficients indicate either conflicting or synergistic relationships.

To explore how structural differences shape these interaction patterns, we further grouped carbon pricing policies according to key design characteristics, including price levels, emission coverage and sectoral scope. For carbon tax regimes, a data-driven *K*-means clustering approach identified two broad implementation patterns: systems focused on energy and industrial sectors, and multi-sector systems characterized by relatively lower policy intensity. Carbon taxes

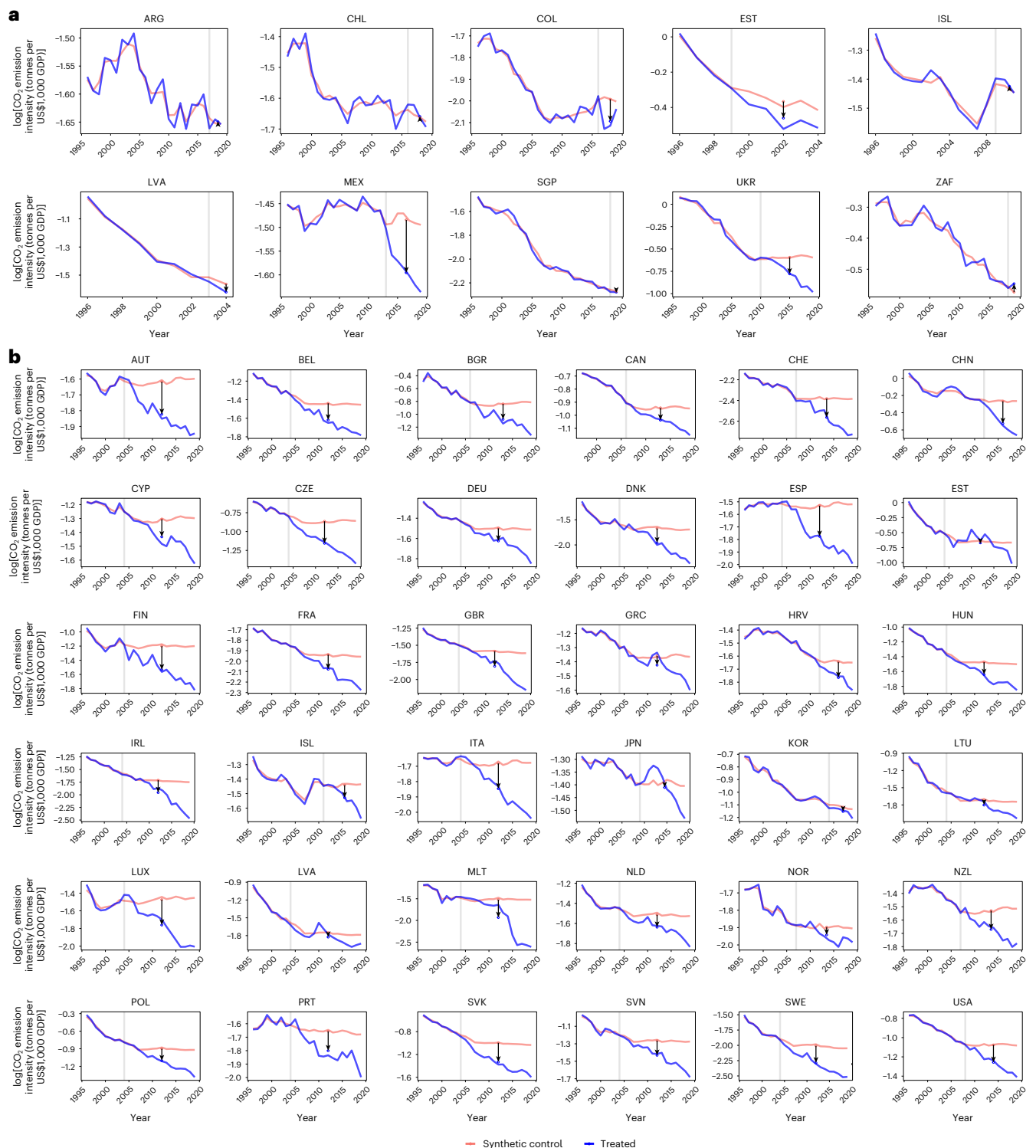


Fig. 1 | The treatment effects of carbon taxes and ETS policies in different countries. **a**, Carbon tax effects: countries that introduced carbon taxes before 1996 (Denmark, Finland, Norway, Poland, Slovenia and Sweden) were excluded because they lack sufficient pretreatment periods to construct valid counterfactuals. To avoid overlapping treatments, eight countries that adopted carbon taxes only after implementing ETSs were excluded from the analysis. **b**, ETS effects: Australia was excluded due to the short-lived nature of its carbon trading scheme (2012–2014). In **a** and **b**, the y axes denote the logarithm of CO₂ emission intensity (tonnes per US\$1,000 gross domestic product (GDP), equivalent to kg CO₂ per US\$ of GDP). Each sub-graph presents the results for an individual country. Abbreviations of country names are defined in Supplementary Table 18. The black arrows indicate the direction of the policy effect, while the vertical grey lines denote the policy adoption time. Baseline emission intensity

levels differ widely across countries, reflecting variations in industrial structure, energy mix and technological development. We allowed intercepts when constructing the synthetic control to ensure parallel pretreatment levels across groups but removed them in the presentation to better illustrate the similarity in pretreatment trends between the synthetic and treated units. During the pretreatment period for each sample, the overlap of two lines suggests that the synthetic control method accurately replicates the emission trends before policy implementation. Although synthetic trajectories appear visually flat due to individualized y-axis ranges, they capture gradual decarbonization consistent with natural trends. Systematic validation confirmed appropriate counterfactual construction through natural trend analysis, pretreatment matching and treatment effect assessment across country types, as well as additional robustness checks and placebo tests (Supplementary Section B).

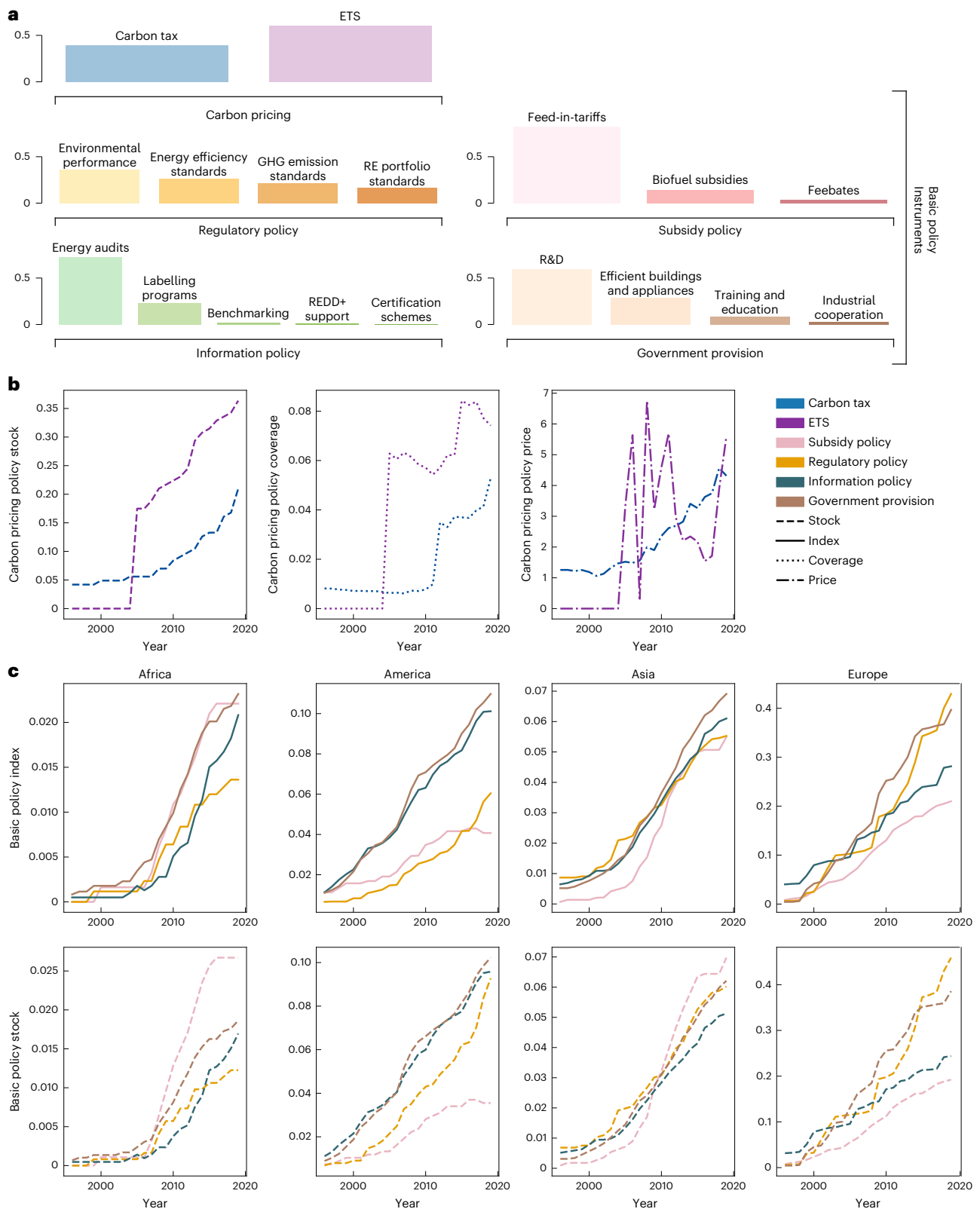


Fig. 2 | National climate change mitigation policy progress during 1996–2019.

The Global Climate Policy Index preserves the overall trend of policy stock growth but further captures underlying heterogeneity through coverage and enforceability weighting, allowing systematic comparison across nations.

a, Composition of all climate change mitigation policy instruments. The bars in each histogram represent the proportion of each policy instrument within the respective category. RE, renewable energy. **b**, Global evolution of carbon taxes and ETSs, showing annual averages of policy numbers as policy stocks (counts), coverage (aggregate share of global emissions) and prices (US\$ per tonne of CO₂).

The data in **b** show the global annual averages of the number, coverage and price of carbon pricing policies, with coverage being the global aggregate.

c, Trends of four basic policy instruments in 124 countries from 1996 to 2019, based on the Global Climate Change Mitigation Policy Dataset. The figure shows the development trends of global climate mitigation policies by continent (Africa, Asia, the Americas and Europe) from 1996 to 2019, analysed using policy indices and stock. The carbon pricing policy index is constructed as price × coverage. The data in **c** have been normalized.

create a uniform price signal across emitters, but their effectiveness depends critically on how broadly that signal reaches and how governments recycle the revenues. The regression results show clear synergies between carbon taxes and government provision policies, particularly in multi-sector systems with relatively low tax rates, where a 0.01 increase in the interaction term yields a statistically significant reduction in emission intensity of 1.078% (Extended Data Table 1, column 3; $P < 0.05$). Even at modest tax levels, broad coverage extends the pricing signal across the economy, increasing the scope of potential abatement. By contrast, when carbon taxes are introduced after ETSs, interactions with subsidy policies frequently weaken mitigation outcomes.

The effectiveness of ETS interactions is strongly shaped by whether complementary policies support market functioning and price formation or create competing signals. We also considered the heterogeneous designs of ETSs, such as market scheme, price and allowance allocation. In particular, the EU ETS is distinguished from others due to its broader sectoral and geographic coverage, while non-EU ETSs are grouped by allowance allocation rules (mass-based versus rate-based) and overall policy intensity.

Both the mass-based ETSs and the EU ETS exhibit strong conflicts with subsidy policies. Despite differences in participating countries, these systems share a common design feature—a quantity-based cap on total emissions. In theory, a perfectly functioning ETS operates under a fixed emissions cap, allowing the market to determine the cost-effective allocation of abatement through price signals^{47,48}. Subsidizing renewable energy generates dual opposing effects. On the positive side, it directly reduces emissions by displacing fossil-fuel-based power generation⁴⁹. On the negative side, it lowers demand for emissions allowances, thereby depressing the carbon price, potentially undermining the ETS's allocative efficiency and weakening incentives for broader mitigation⁵⁰. Additional rebound effects—such as lower electricity prices stimulating higher demand—may further weaken system-wide mitigation outcomes. However, in lower-intensity ETSs with relatively limited coverage (for example, China before 2021, where only pilot regions were included), different interaction patterns emerge⁵¹. In these contexts, nationwide renewable subsidies tend to generate synergistic effects, as direct abatement in uncovered sectors can outweigh distortions in allowance markets, as shown in column 11 of Extended Data Table 1. Nonetheless, such subsidies should be gradually phased out as ETS coverage and market strength improve and ensure long-term cost-effective reductions.

Information and regulatory policies display heterogeneous associations across ETS types. Within the EU ETS, information instruments and regulatory standards tend to align with stronger mitigation outcomes, reflecting the integration of monitoring, reporting and verification systems into a broader institutional framework⁵². To mitigate carbon leakage risks among multinational firms, the EU mandates emission reporting and rigorous third-party verification of annual allowances. Such regulatory instruments complement the carbon price signal by addressing information asymmetries and guaranteeing minimum emission standards^{33,53}. However, in less developed or low-intensity ETSs, information programmes tend to induce conflict patterns, as firms may prioritize symbolic compliance over cost-effective abatement. These non-market responses can distort the price signal, weakening the abatement efficiency. Limited administrative capacity and data infrastructure further constrain the implementation of effective monitoring, reporting and verification in these settings, contributing to weaker alignment with price signals^{54–57} (Extended Data Table 1, columns 8 and 11).

Meanwhile, in low-intensity ETSs, carbon pricing policies exhibit strong synergies with government provision policies. By supporting research and development and public infrastructure, such policies reduce compliance costs within the ETS cap and provide an early boost to decarbonization⁵¹. Although implementing this type of government policy supporting research and development alone is inefficient³³, it

can address externalities of low-carbon technologies more moderately, serving as an effective supplementary tool for ETS and assisting in the long-term transformation path at a lower social cost.

These findings highlight the critical need to untangle policy interactions in relation to the specific characteristics of carbon markets, and we summarize the underlying mechanisms in Supplementary Section A. Aligning subsidy and information policies with ETS maturity, while leveraging government provision instruments to ease adjustment costs, promotes more coherent mitigation outcomes⁵⁸. An event study analysis demonstrates the emission reduction patterns of grouped ETSs (Fig. 3). Interactions involving government provision policies play a major role in synergistic mitigation effects, especially in low-intensity ETSs. By contrast, subsidies—such as feed-in tariffs for renewable electricity, electric vehicle purchase subsidies and clean technology investment incentives—tend to conflict with most ETS objectives. This analysis is further extended by decomposing emission reduction patterns linked to other basic policy instruments in Supplementary Section B. Overall, these patterns reinforce existing theoretical and empirical insights and highlight the central role of policy design and interaction in shaping carbon pricing outcomes^{24,49,59,60}.

Counterfactual analysis on climate policy selection

Successful mitigation of carbon emissions can be further enhanced by improving policymaking to reduce policy conflicts. We thus simulated the potential benefits of reducing policy conflict levels in eight groups of carbon pricing samples (Fig. 4). Rather than assuming policy removal, these scenarios represent alternative configurations in which overlapping instruments are better aligned or implemented with lower intensity. Specifically, aggregated conflict levels were reduced by 20%, 40%, 60%, 80% and 100%, and we recalculated the corresponding emission-intensity reductions on the basis of the regression coefficients (Extended Data Table 1). The results show that improving the coherence between carbon pricing and complementary policies can substantially improve mitigation outcomes. Under scenarios in which conflicts are fully neutralized, average emission-intensity reductions increase to nearly 22.3%. For countries implementing carbon taxes, eliminating conflicting overlaps could yield about 1 Gt of additional abatement, whereas the largest relative gains occur in low-intensity ETSs, where a 20% reduction in policy conflicts leads to an extra 5 GtCO₂.

We also separately assessed emission reductions linked to complementary policy instruments themselves (Supplementary Table 4). These policies—such as subsidies—indeed contribute to emission reductions, though their observed effects are considerably smaller than those of carbon pricing. Moreover, these gains may be partially offset by the conflict with carbon pricing policies, reinforcing the importance of policy alignment.

These simulated outcomes do not imply that removing emission-reducing policies would lower emissions. Instead, they illustrate efficiency gains arising from better coordination among instruments. In mature ETSs, greater coherence may help prevent subsidies from weakening price signals, whereas in early-stage systems, targeted support can continue to complement carbon pricing. Overall, these results emphasize that strengthening policy coherence, rather than eliminating specific policy tools, can substantially increase the effectiveness of carbon pricing in deep decarbonization pathways.

We further analysed policy selections for countries that had not yet implemented carbon pricing policies as of 2019. For each country, the most suitable carbon pricing policy was identified according to the characteristics of its basic policies. On the basis of the regression results (Extended Data Table 1), we simulated the implementation of various types of carbon pricing policies in 94 countries and assessed how policy interaction patterns relate to their emission intensity. The policy option that resulted in the lowest simulated emission intensity was selected as the most appropriate for each country

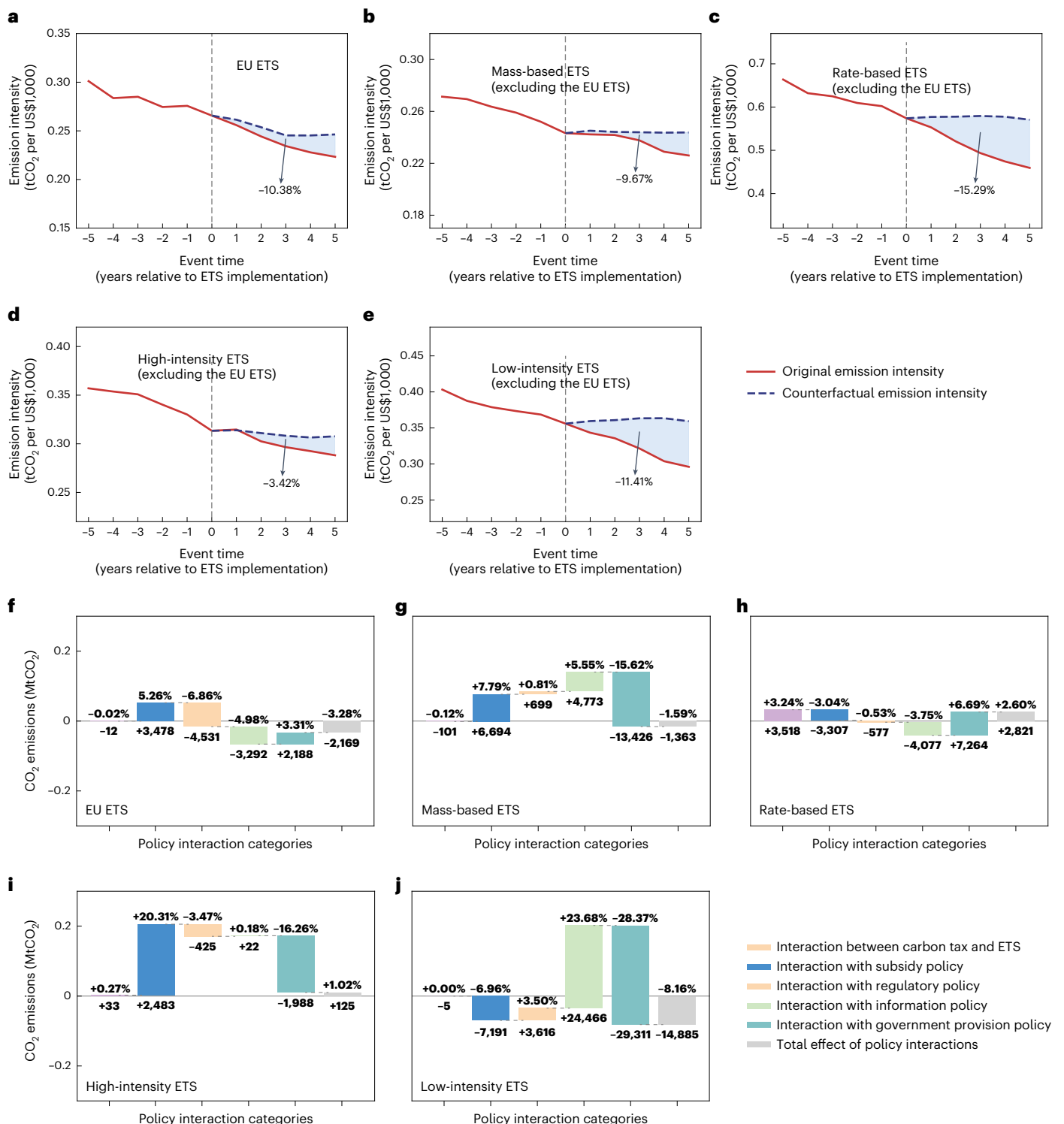


Fig. 3 | Direct and decomposed policy effect and interactions on ETSs.

a–e, The event study of treatment effects from ETSs, including the time-fixed effects. Original and counterfactual emission intensity trajectories depict realized and hypothetical emission paths following policy implementation, respectively. Since there are overlaps between the carbon tax and trading, a five-year event study result is presented to showcase the policy effect. Hence, the reduction percentage is slightly different from the full sample. The year when a country first implemented carbon pricing policies is set as the first year and the previous year as the zeroth period. The values in the figure represent the

change in the weighted average of emission intensity for all periods with and without policies, thus reflecting the aggregated effect. **f–j**, The changes in CO₂ emissions caused by each interaction effect, except for the time effect. Each bar in the figures indicates the magnitude of the change in CO₂ emissions caused by the interaction of the corresponding policies during 1996–2019, with the corresponding percentage and absolute emission (in parentheses, measured in million tonnes) values below. The grey bars represent the sum of all interaction effects in each group. The treatment effects and the decomposed interactions of the three groups of carbon tax policies can be found in Supplementary Fig. 7.

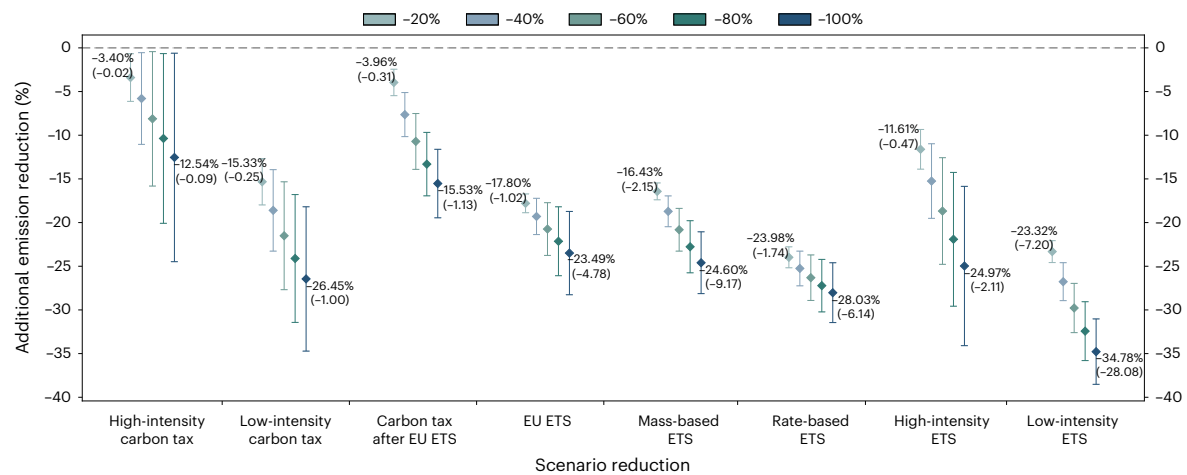


Fig. 4 | Policy effect simulation under different conflict policy scenarios.

Results of a counterfactual policy simulation designed to estimate the potential emission reduction gains from decreasing the level of conflict between carbon pricing instruments and other climate policies. For each of the eight carbon pricing subsamples, the intensity of policy conflicts—measured by an aggregated conflict index based on the interaction terms—was systematically reduced by 20%, 40%, 60%, 80% and 100%. Predicted changes in CO₂ emission intensity were calculated on the basis of the regression coefficients estimated from the baseline models. Values above the markers indicate the additional

percentage in emission reductions, and the numbers in parentheses indicate the corresponding additional absolute emission reductions (GtCO₂). The error bars represent 90% confidence intervals, derived from Monte Carlo simulations with 10,000 draws from the estimated variance–covariance matrix of the regression coefficients. The unit of analysis is the country–year observation, with n equal to the number of country–year observations in each subsample, as reported in Supplementary Table 20. Uncertainty reflects parameter uncertainty from the regression estimates.

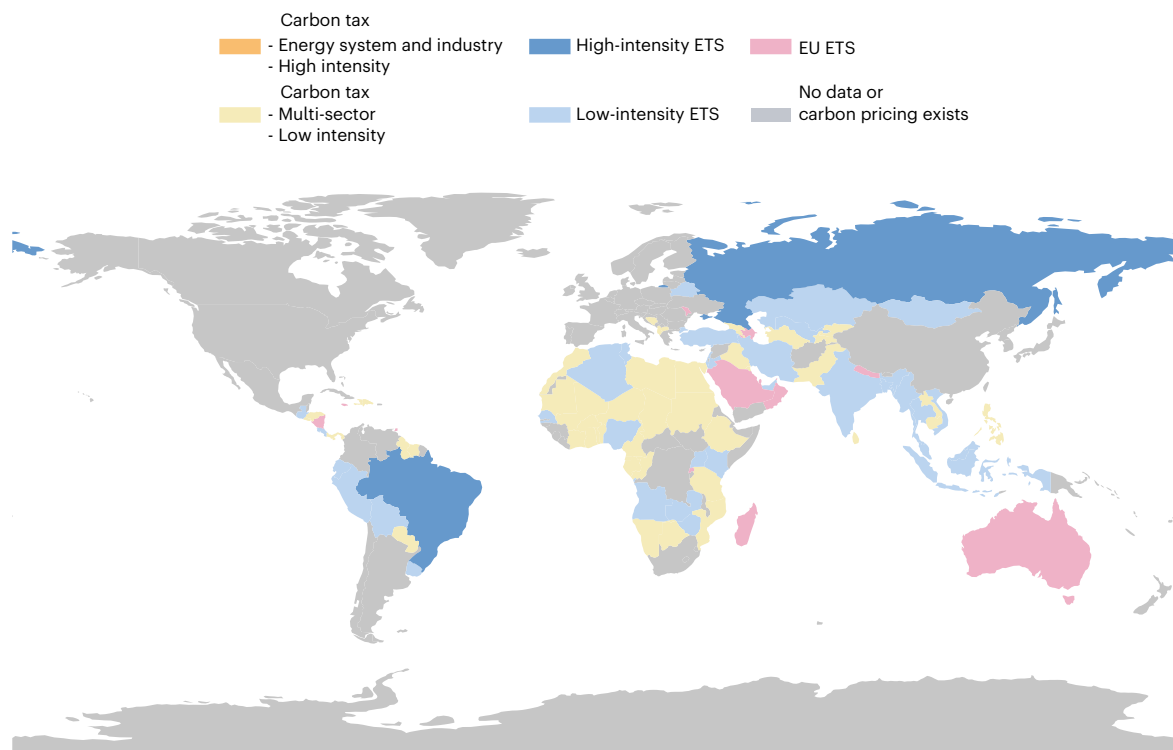


Fig. 5 | Counterfactual analysis and climate policy selection. Simulated carbon pricing policy choices for countries without carbon pricing policies based on the basic policy index in 2019. Policy choices were determined by comparing the emission intensity of each country under different policies.

Colours indicate the policy chosen in each country. Source data for this figure are available in Supplementary Table 21. Basemap from Natural Earth (<https://www.naturalearthdata.com>).

(Fig. 5). The simulation results indicate that 12 countries are suitable for the EU ETS, 34 for low-intensity ETSs, and only Brazil and Russia for high-intensity ETSs. Additionally, 45 countries, mostly with limited basic policies, are better suited for low-intensity, multi-sector carbon taxes (Supplementary Table 21).

The interaction patterns between basic policies and carbon pricing policies highlight that these relationships vary across different types of carbon pricing. Before implementing carbon pricing policies, countries should assess potential conflicts or synergies with their existing policy foundations and may strengthen their basic policy

framework to enhance policy mix synergies. The simulation results suggest that some countries, such as Australia and Saudi Arabia, are better suited to implementing an EU ETS-style carbon ETS. These countries' strong regulatory and information policy frameworks can create synergistic effects in emission reduction, making this policy more effective than other types of carbon pricing mechanisms. Successful cases in some countries cannot be straightforwardly replicated in others, and the interaction patterns between policies should be thoroughly considered.

Discussion

Climate policies play a crucial role in reducing carbon emissions worldwide. Understanding the synergies and conflicts among policy instruments is essential for designing effective mitigation strategies, as policies with similar frameworks can produce markedly different outcomes across institutional and economic contexts. For countries already pursuing decarbonization, optimizing policy portfolios can enhance efficiency and reduce mitigation costs. In contrast, countries currently without carbon pricing policies need to effectively design and adjust their existing policy schemes before adopting large-scale measures.

These findings reveal that the policy interaction patterns are, if not the key causal factor, at least closely associated with the performance of both carbon taxes and ETSs. This analysis suggests that the alignment between carbon pricing and complementary instruments varies systematically across policy types. For example, multi-sector carbon taxes may be more effective when combined with government-provided policies that promote the development of green technologies. ETSs often exhibit strong synergies with regulatory policies in promoting emission reductions, with more heterogeneous relationships involving information and subsidy policies. These patterns highlight the importance of considering policy interactions when designing coherent and context-sensitive climate strategies. By considering local policy contexts and interaction patterns, scenario-based simulations highlight how different carbon pricing designs align with existing national policy foundations. This approach accounts for the unique climate policy conditions of each country, showing that a one-size-fits-all strategy is insufficient. A well-designed, interactive policy portfolio is crucial for effective climate change mitigation.

Moreover, decarbonization rarely hinges on a single, comprehensive measure. Instead, it evolves incrementally, with earlier measures shaping the context in which subsequent policies operate^{61–63}. An event study analysis investigates the dynamic effects of basic climate policies with either ETS or carbon tax regimes (Supplementary Section C1), providing additional evidence for the claims regarding policy interaction and layering.

Overall, the results highlight the need for a dynamic approach to policy evaluation, as similar policy mixes can be associated with divergent outcomes depending on contextual factors and the institutional maturity of carbon markets. Patterns linking renewable energy development and carbon pricing performance suggest that higher renewable penetration tends to align with stronger mitigation outcomes, particularly in low-intensity pricing systems (Supplementary Section C4). These results underscore the importance of adaptive and coordinated policy design over the accumulation of overlapping instruments. Emerging economies may benefit from combining carbon pricing with targeted complementary measures, while advanced economies may prioritize policy streamlining, regulatory coherence and cross-border coordination. More broadly, the coherence and credibility of national policy portfolios appear more informative than carbon prices alone in international climate governance.

A broad set of robustness checks, detailed in the Methods and Supplementary Information, supports the stability of these findings across alternative specifications. These include placebo tests, expanded covariate balancing, partial identification exercises, spillover and

leakage adjustments, and sensitivity analyses of policy index construction. While endogeneity cannot be fully ruled out, the convergence of results across approaches lends confidence to the policy-relevant patterns identified.

While this study provides policy-relevant insights into how climate policy interactions shape mitigation outcomes, several limitations warrant caution. First, endogeneity cannot be fully ruled out in a global observational setting, despite extensive robustness checks. Future research may explore the use of instrumental variable strategies to further strengthen causal identification⁶⁴. Second, cross-country differences in policy enforcement and public acceptance may lead to heterogeneous outcomes even under similar policy frameworks. Third, although this paper focuses on the heterogeneous effects of carbon pricing and provides a helpful blueprint for policy portfolios, it does not fully explore the richness of other basic mitigation policies. Fourth, this analysis focuses on domestic and production-based emission intensity, aiming to capture how national carbon pricing and policy interactions affect domestic abatement outcomes. However, these estimates do not fully reflect global mitigation effects, as domestic reductions may be partly offset by carbon leakage through trade and production relocation. Future research combining multi-region input–output data with cross-border policy diffusion models could provide a more integrated assessment of global emission displacement and policy coordination⁶⁵.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41558-026-02574-4>.

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Methods

Data

This study used data from the Global Climate Change Mitigation Policy Dataset⁴⁴, which harmonizes and extends several leading climate policy databases (for example, the International Energy Agency, the Climate Policy Database and Climate Change Laws of the World) and documents over 10,000 climate mitigation policies across more than 100 countries. The dataset provides standardized information on policy instruments, sectoral coverage and legal status (Supplementary Table 18). There have been three distinct surges in policy adoption (Supplementary Fig. 13), occurring in 1997 (the Kyoto Protocol), 2009 (the Copenhagen Conference) and 2015 (the Paris Agreement).

Information on carbon pricing policies—including annual price levels and emissions coverage—was compiled from the World Bank Carbon Pricing Dashboard and the World Carbon Pricing Database. Climate policies were classified into carbon pricing instruments and four categories of complementary policies following IPCC standards⁶⁶. Policy strength, jurisdictional scope and legal enforceability were incorporated into index construction. Detailed classification criteria, legal weighting schemes and robustness checks are reported in the Supplementary Information.

National emission intensity was constructed using sector- and sub-sector-level CO₂ emissions data from the Emissions Database for Global Atmospheric Research⁶⁷, with missing values interpolated using adjacent-year averages.

The Global Climate Policy Index

The Global Climate Policy Index is structured around two components: carbon pricing policies and four basic climate policy instruments capturing subsidies, regulations, information policies and government provision.

The carbon pricing policy index

For carbon pricing policy, we used the price and the proportion of emissions covered. The carbon pricing policy index $CPI_{it,n}$ is constructed as follows:

$$CPI_{it,n} = Price_{it,n} \times Coverage_{it,n} \quad (1)$$

where $Price_{it,n}$ and $Coverage_{it,n}$ are the average prices (US\$ per tonne) and emissions coverage ratios (%) of the carbon pricing policy of country i in year t . This formula was applied separately to the two types of carbon pricing policies ($n = 2$): carbon taxes and ETSs. Specifically, $CPI_{it,1}$ was calculated for carbon tax policies (33 cases), and $CPI_{it,2}$ was calculated for ETS policies (28 cases) implemented as of 2019. In countries with multiple ETS markets (for example, China's eight pilot carbon markets or Canada's regional and national markets), the annual ETS index was obtained by aggregating the indices of these markets. The specific characteristics of carbon pricing policy can be found in Supplementary Figs. 14–18.

To systematically capture heterogeneity in carbon pricing design across countries, we employed a machine learning clustering approach. Policies were classified into eight distinct groups. For carbon tax policies, the K -means clustering method was employed, with input features including the average price, the sectors regulated and the emissions covered. This yielded two groups: (1) covering energy systems and industrial sectors with high intensity, and (2) covering multiple sectors (including construction, transportation, energy systems and industrial sectors) with low intensity. Additionally, carbon taxes adopted after the introduction of the EU ETS were treated separately to reflect differences in policy sequencing. The K -means algorithm identifies latent policy archetypes in a high-dimensional feature space, allowing policy groupings to emerge endogenously from observed design characteristics.

The EU ETS was classified into a separate group to account for potential differences in policy dynamics. ETSs were categorized

according to market design and policy intensity. The EU ETS was treated as a distinct group due to its extensive sectoral scope, geographic coverage and institutional integration. Other ETSs were distinguished by cap design (mass-based versus rate-based) and further classified into high- and low-intensity systems on the basis of CPI values. This classification enables systematic comparison across diverse carbon pricing architectures. The detailed grouping results are reported in Supplementary Figs. 19–21.

The basic policy instruments of climate change mitigation

To assess the implementation of basic climate change mitigation policies, we compiled and calculated the number of existing policies across different countries and regions, constructing a policy stock index following the methodology of Chen et al.⁶⁸. As for the measurement of policy index, the concepts of “effectiveness”, “importance” and “strictness” have been repeatedly mentioned in relevant literature^{69,70}. Some scholars have proposed to measure policy intensity by combining the policy objectives, scope, implementation and other indicators^{71,72}. The importance and effectiveness of climate policies can be proxied through their legal status—specifically, whether they take the form of hard law or soft law. Hard laws require extensive inter-ministerial coordination and formal legislative processes, providing stronger legal enforceability but higher adoption costs. Soft laws are more flexible but may have limited follow-through. On the basis of these foundations, we constructed the comprehensive policy index for basic climate change mitigation instruments as follows:

$$BPI_{it,m} = \sum_l \sum_k HS_{i,l,m,k} \times JU_{i,l,m,k} \quad (2)$$

where $BPI_{it,m}$ are the m th basic climate mitigation policy indices (subsidy, regulatory, information and government provision policies) for country i in year t . $HS_{i,l,m,k}$ are the hard law and soft law weighted scores of policy instrument k within policy m of country i in year l , and $JU_{i,l,m,k}$ are the weighted scores of jurisdictions. A climate change mitigation policy usually includes one or more policy instruments; for example, regulatory policies include environmental performance standards, energy efficiency standards for equipment, GHG emission standards and renewable portfolio standards (for example, $m = \text{Regulatory}$, $k = 4$). Due to a lack of data, we did not consider the validity period of the policy, and we assumed that once a policy is promulgated, it will remain in effect. The index of a country's policy m in year t is therefore a weighted calculation of all the policy instruments distributed between l and t , where l is the year when basic policy m first appeared.

Legal enforceability distinguishes between soft and hard law, while jurisdictional scope differentiates international, national and subnational policies. Detailed legal classifications, weighting schemes and policy definitions are provided in Supplementary Section D3 and Supplementary Tables 22–24.

We implemented a series of robustness checks using alternative weighting schemes (Supplementary Tables 27–59). We also constructed a policy stock index that omits legal and jurisdictional weights entirely, relying solely on the number of adopted policies (Supplementary Table 60). Across all specifications, the key findings remained robust, indicating that the overall patterns of policy interaction were not driven by any single set of weight assumptions. Furthermore, we tested the impact of subnational policies by excluding them from the index (Supplementary Table 61).

The ‘two-step’ regression method

At present, research on policy interactions, particularly synergies, remains relatively nascent. Existing studies primarily focus on simulating trade-offs between multiple objectives, such as climate and environmental goals, climate and sustainable development goals^{73,74}, different sustainable development goals⁷⁵, mitigation and adaptation

goals⁷⁶, climate change mitigation and national development goals⁷⁷, and social and ecological goals⁷⁸. Existing methods for studying policy interactions can be broadly classified into three categories: (1) case studies, which analyse the synergy between policy tools and policy design in one or several countries^{27,28,79,80}; (2) literature reviews, which conduct qualitative analysis by sorting out the research results of different literature^{8,81,82}; and (3) theoretical modelling or simulation to analyse synergy between two or more policy tools^{31–35}.

Many scholars have used quantitative analysis to demonstrate the effectiveness of carbon pricing policies²², with most studies relying on the DID method. However, the DID method faces two key challenges: first, the selection of the reference group is subjective and often unconvincing, and second, policy endogeneity creates systematic differences between countries that implement carbon pricing and those that do not, as these differences may influence the adoption of such policies. Given these issues, DID estimates are prone to bias. To address these shortcomings, Abadie and Gardeazabal⁸³ and Abadie et al.⁸⁴ introduced the synthetic control method, which improves estimation by constructing a weighted combination of reference groups, avoiding subjective selection.

The synthetic model

We first treated the carbon tax or trading as ‘treatments’. We then applied the synthetic method country by country, using countries that never adopted such policies to construct a synthetic counterfactual time trend of emission intensity. Specifically, we constructed a ‘synthetic pool’ of countries that have never implemented any carbon pricing policies and synthesized the counterfactual emission intensity for each country that introduced carbon tax or ETS policies between 1996 and 2019. The detailed model can be written as:

$$y_{i,t} = \alpha_i + \sum_j^N y_{j,t}^0 w_{ij} \text{ when } t < t_0 \quad (3)$$

where, $y_{i,t}$ represents the logarithmic form of the emission intensity for country i in year t . We then used the emission intensity $y_{j,t}^0$ of countries in the synthetic pool that never adopted carbon pricing, with a weighting vector w_i (the difference across each target country i) to construct a synthetic sample with the least distance to the real curve; t_0 is the time point at which country i conducted the relative climate policy, either tax or trading, and N^0 indicates the number of countries without carbon pricing policies. As climate policies might be concentrated in countries with lower emission intensity, we allowed a certain intercept α_i between the synthetic intensity and the real one⁸⁴. We also provide the synthesized weight matrix for each country in Supplementary Tables 62–64.

To estimate the carbon policy effect $\tau_{i,t}$ for country i in year t , we calculated the difference between the observed and counterfactual emission intensity:

$$\tau_{i,t} = \Delta y_{i,t} = y_{i,t} - \widehat{y}_{i,t} \text{ when } t \geq t_0 \quad (4)$$

where the counterfactual $\widehat{y}_{i,t}$ is constructed as:

$$\sum_a^b \widehat{y}_{i,t} = \alpha_i + \sum_j^N y_{j,t}^0 w_{ij} \text{ with } t \geq t_0 \quad (5)$$

This equation estimates the hypothetical emission intensity of country i , assuming no carbon pricing policy had been adopted. The difference $\tau_{i,t}$ thus captures the treatment effect of the policy.

This process allowed us to obtain the treatment effect $\tau_{i,t}$ for each country and each year, corresponding to various policy indices and interactions. We then summarized the treatment effect vector τ_i of each country (timely observations of each country) and formed the dependent variable τ for the following analysis.

The regression model

On the basis of the emission reduction effects generated by the adoption of carbon pricing policies in the previous synthetic model, we estimated the interactions between the different basic policies and carbon pricing policies through the following general interaction regression model:

$$\tau_{i,t,n} = \sum_m \beta_m \text{CPI}_{i,t,n} \times \text{BPI}_{i,t,m} + \mu_{i,t,n} + \varepsilon_{i,t,n} \quad (6)$$

where $\tau_{i,t,n}$ is the emission reduction effect in country i in year t , associated with the carbon pricing policy n . $\text{CPI}_{i,t}$ is the carbon pricing policy indices for country i in year t ; $\text{BPI}_{i,t,m}$ represents the basic policy instrument indices m , including the subsidy policy index ($\text{BPI}_{m=\text{SUB}}$), the government provision policy index ($\text{BPI}_{m=\text{GPP}}$), the information policy index ($\text{BPI}_{m=\text{IP}}$) and the regulatory policy index ($\text{BPI}_{m=\text{RA}}$). The coefficients β_m capture the interaction effects between the m th basic policy instrument and carbon pricing policies. $\mu_{i,t,n}$ is the time-fixed effect, indicating the year sequence after the policy n was implemented, and $\varepsilon_{i,t,n}$ is the error term. When estimating the policy interaction effects on ETS, we covered the interaction term between the carbon tax and ETS policy to see whether climate policies themselves would drive synergistic or conflicting outcomes. It is worth noting that this time-fixed effect might range across different countries and policies since countries adopted different carbon pricing policies in different years.

On the basis of the classification of carbon pricing policies, we conducted a total of 11 regressions, as presented in Extended Data Table 1. These regressions included two full-sample regressions for carbon taxes and ETS policies and nine subgroup regressions based on policy classification. Two specific cases warrant special attention. First, in terms of carbon tax policy, Denmark, Finland, Norway, Poland, Slovenia and Sweden implemented carbon tax policies in 1996 or earlier, leaving no untreated period for constructing a synthetic control. Consequently, these countries were excluded from the carbon tax policy regressions. Second, in the regression of ETS policies, Australia’s ETS, implemented in 2012 and terminated in 2014, was excluded from the analysis due to its short duration, which could distort the estimation of policy effects.

When conducting the regression analysis, we normalized all variables related to the basic policy indices, as follows: $\text{BPI}_{i,t,m} = \frac{\text{BPI}_{i,t,m}^{\text{orig}}}{\text{BPI}_m^{\text{max}}}$, where $\text{BPI}_{i,t,m}^{\text{orig}}$ represents the original m policy index of country i in year t , and $\text{BPI}_m^{\text{max}}$ represents the maximum m policy index of all countries. For detailed variable descriptions and descriptive statistics, see Supplementary Tables 25 and 26.

Robustness and sensitivity

We conducted a comprehensive set of robustness and sensitivity analyses to assess the stability of the main findings. These included placebo tests within the synthetic control framework (Supplementary Fig. 2), alternative specifications incorporating additional economic and institutional covariates (Supplementary Section C2.1) and partial identification exercises exploiting external policy exposure and policy diffusion patterns (Supplementary Section C2.2). We further examined the sensitivity of the results to potential spillovers, spatial dependence and carbon leakage using donor-pool restrictions, spatially robust estimators and consumption-based emissions measures (Supplementary Section C3). In addition, we tested the construction of the Global Climate Policy Index under alternative weighting schemes, and we pooled carbon taxes and ETSs into a unified carbon pricing category as a sensitivity check. The weighted sensitivities can be found in Supplementary Tables 28–32. The results of the robustness checks can be found in Supplementary Tables 33–61. Across all specifications, the core interaction patterns remain stable. The detailed procedures are reported in Supplementary Section D4.

The counterfactual method

To evaluate the impact of carbon pricing policies on CO₂ emission intensity, we performed a counterfactual simulation to estimate what the emission intensity would have been if these policies had not been adopted. This approach follows the framework established in previous research^{12,68}. The counterfactual CO₂ emission intensity was calculated as follows:

$$\ln(\widehat{\text{CO}_2/\text{GDP}})_{i,t} = \widehat{y}_{i,t} = y_{i,t} - \Delta y_{i,t} = \ln(\text{CO}_2/\text{GDP})_{i,t} - \tau_{i,t} \quad (7)$$

where $(\text{CO}_2/\text{GDP})_{i,t}$ is the actual CO₂ emission intensity of country i in year t , and $(\widehat{\text{CO}_2/\text{GDP}})_{i,t}$ represents the counterfactual emission intensity. Here, $\tau_{i,t}$ denotes the treatment effect of carbon pricing policies estimated from the interaction regression model. To generate the counterfactual emission intensity curves, we grouped countries according to their adopted carbon pricing policies and averaged the counterfactual emission intensity estimates within each group over the treatment period.

Decomposed interaction effect on emission

This decomposition allowed us to quantify the relative importance of different policy interactions and their contribution to emission reduction outcomes. The contribution of each interaction effect to emission reductions was calculated as follows:

$$\text{prop}_{t,m}^p = \beta_m^p \text{CPI}_{it}^p \times \text{BPI}_{it,m}^p / \Delta y_{it}^p \quad (8)$$

where $\beta_m^p \text{CPI}_{it}^p \times \text{BPI}_{it,m}^p$ represents the interaction effect between policies in the p th group (the eight groups of carbon pricing policies), and Δy_{it}^p is the total emission reduction effect brought about by the policy. By averaging $\text{prop}_{t,m}^p$ across treatment periods for all countries within each group, we obtained the proportion of the impact of each interaction term (prop_m^p) in the overall emission reduction effect.

The emission reduction attributable to each interaction term was then calculated by multiplying its contribution proportion by the total emission reduction within each group:

$$\text{reduction}_m^p = \text{prop}_m^p \times \sum_t \sum_i (\text{CO}_2 - \widehat{\text{CO}_2})_{it}^p \quad (9)$$

where CO_2 is the real emission, $\widehat{\text{CO}_2}$ is the counterfactual emission and $\sum_t \sum_i (\text{CO}_2 - \widehat{\text{CO}_2})_{it}^p$ denotes the total emission reduction for group p . This calculation allowed us to estimate the emission reduction (reduction_m^p) attributable to each interaction term within different groups of carbon pricing policies. By decomposing the interaction effects, we could quantify the contribution of each policy interaction to the total emission reductions across different groups. This approach not only highlights the relative importance of specific policy combinations but also provides a more detailed understanding of how carbon pricing policies interact with basic policy instruments to achieve emission reduction targets.

Policy effect simulation with different conflict policy scenarios

To assess the impact of reducing conflicting basic policies on the emission reduction effects of carbon pricing policies, we simulated policy effects under different conflict reduction scenarios. We defined the adjusted policy effect as $\widehat{\Delta y}_{it,k}^p$, calculated by reducing the contribution of basic policies that conflict with carbon pricing policies. The adjusted policy effect is expressed as follows:

$$\widehat{\Delta y}_{it}^p = \Delta y_{it}^p - \sum_{m \in \text{conflicts}} \beta_m^p \text{CPI}_{it}^p \times \lambda \text{BPI}_{it,m}^p \quad (10)$$

where $\text{BPI}_{it,m}^p$ represents the basic policy index that conflicts with the carbon pricing policy in group p , and $\lambda = 0.2, 0.4, 0.6, 0.8$ and 1.0 denotes five levels of conflict reduction scenarios.

Using the emission intensity without carbon pricing policies, estimated earlier, we calculated the adjusted emission intensity under different scenarios for group p as follows:

$$\ln(\widehat{\text{CO}_2/\text{GDP}})_{i,t}^p = \ln(\widehat{\text{CO}_2/\text{GDP}})_{i,t}^p - \widehat{\Delta y}_{i,t}^p \quad (11)$$

where $(\widehat{\text{CO}_2/\text{GDP}})_{i,t}^p$ represents the counterfactual emission intensity under different conflict reduction scenarios, and $(\widehat{\text{CO}_2/\text{GDP}})_{i,t}^p$ represents the emission intensity without adopting carbon pricing policies.

To quantify the uncertainty in the scenario-based simulations, we applied a Monte Carlo simulation with 10,000 draws from the estimated variance–covariance matrix of regression coefficients. This yielded 90% confidence intervals for the predicted effects under different degrees of policy conflict reduction (20%, 40%, 60%, 80% and 100%). The full numerical simulation outputs are reported in the Supplementary Table 20.

Counterfactual analysis of climate policy selection

To further explore the potential impacts of carbon pricing policies, we conducted a counterfactual simulation for countries that have not yet implemented any carbon pricing policies. On the basis of the various basic policy indices of these countries in 2019, we calculated the interaction effects between policies and estimated the emission reduction effects of implementing different types of carbon pricing policies. The emission reduction effect is calculated as follows:

$$\widehat{\Delta y}_j^p = \sum_m \beta_m^p \overline{\text{CPI}}_j^p \times \text{BPI}_{j,m} + \overline{\mu}_p \quad (12)$$

where $\widehat{\Delta y}_j^p$ represents the emission reduction effect of implementing a carbon pricing policy of type p in country j , which currently lacks carbon pricing policies. Here, $\text{BPI}_{j,m}$ is the index of the basic policy m of country j as of 2019, and $\overline{\text{CPI}}_j^p$ denotes the average carbon pricing policy intensity for group p . To reflect the long-term effects of implementing carbon pricing policies, we added $\overline{\mu}_p$, which represents the five-year average of time treatment effects derived from the regression results of each group of carbon pricing policies. Using this approach, we estimated the emission reduction effects for eight different types of carbon pricing policies across 94 countries without any carbon pricing policies. By comparing the results, we identified the carbon pricing policy that minimizes emission intensity for each country, thus selecting the most suitable policy option for their specific circumstance. The policy simulation output is reported in detail in Supplementary Table 21.

Data availability

All data used in this study are publicly available. The climate policy data were obtained from the Global Climate Change Mitigation Policy Dataset. The carbon pricing data were compiled from the World Carbon Pricing Database (WCPD) and the Carbon Pricing Dashboard of the World Bank. The processed datasets generated and analysed in this study are publicly available via Zenodo at <https://doi.org/10.5281/zenodo.18015601> (ref. 85). Source data are provided with this paper.

Code availability

A replication package is publicly available via Zenodo at <https://doi.org/10.5281/zenodo.18015601> (ref. 85).

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Author contributions

L.W. and Y.Z. conceived the idea and designed the study. L.W. and J.M. supervised the study. G.L., Z.H. and Y.Z. contributed to the methodology. G.L. and Y.Z. analysed the data and generated the figures. G.L. wrote the first draft of the paper, and all authors provided feedback. L.W., G.L., J.M. and Y.Z. contributed substantially to editing the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

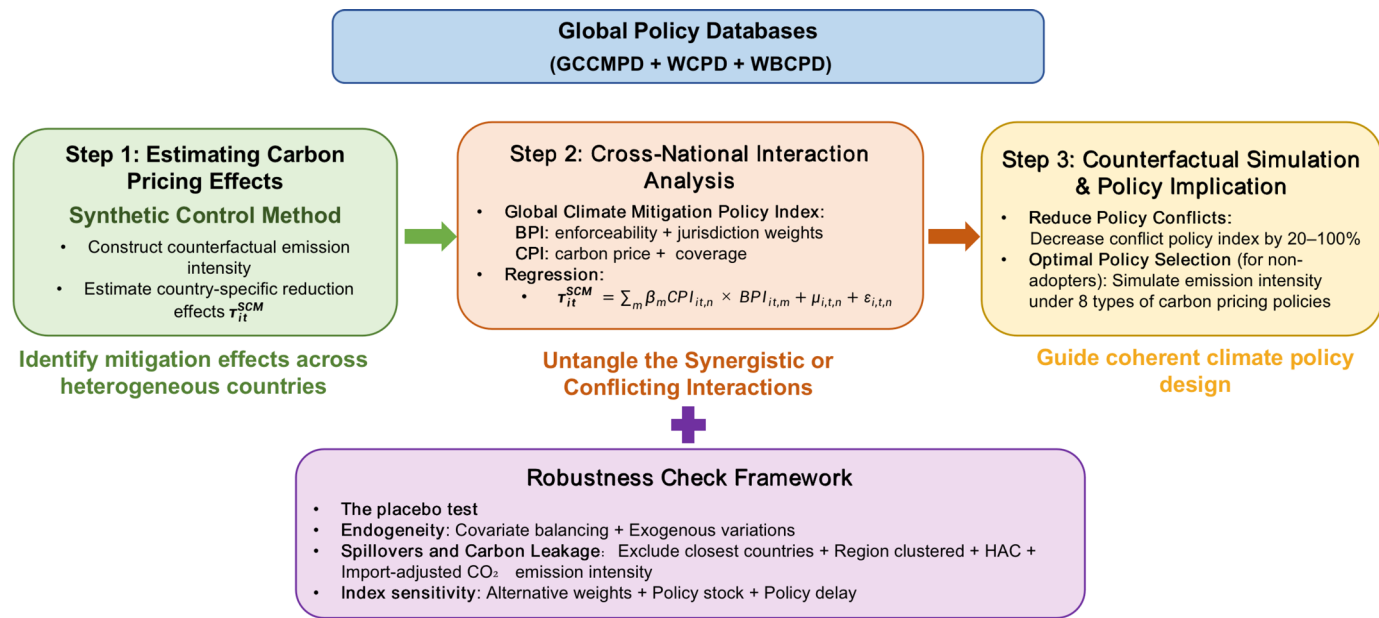
Extended data is available for this paper at <https://doi.org/10.1038/s41558-026-02574-4>.

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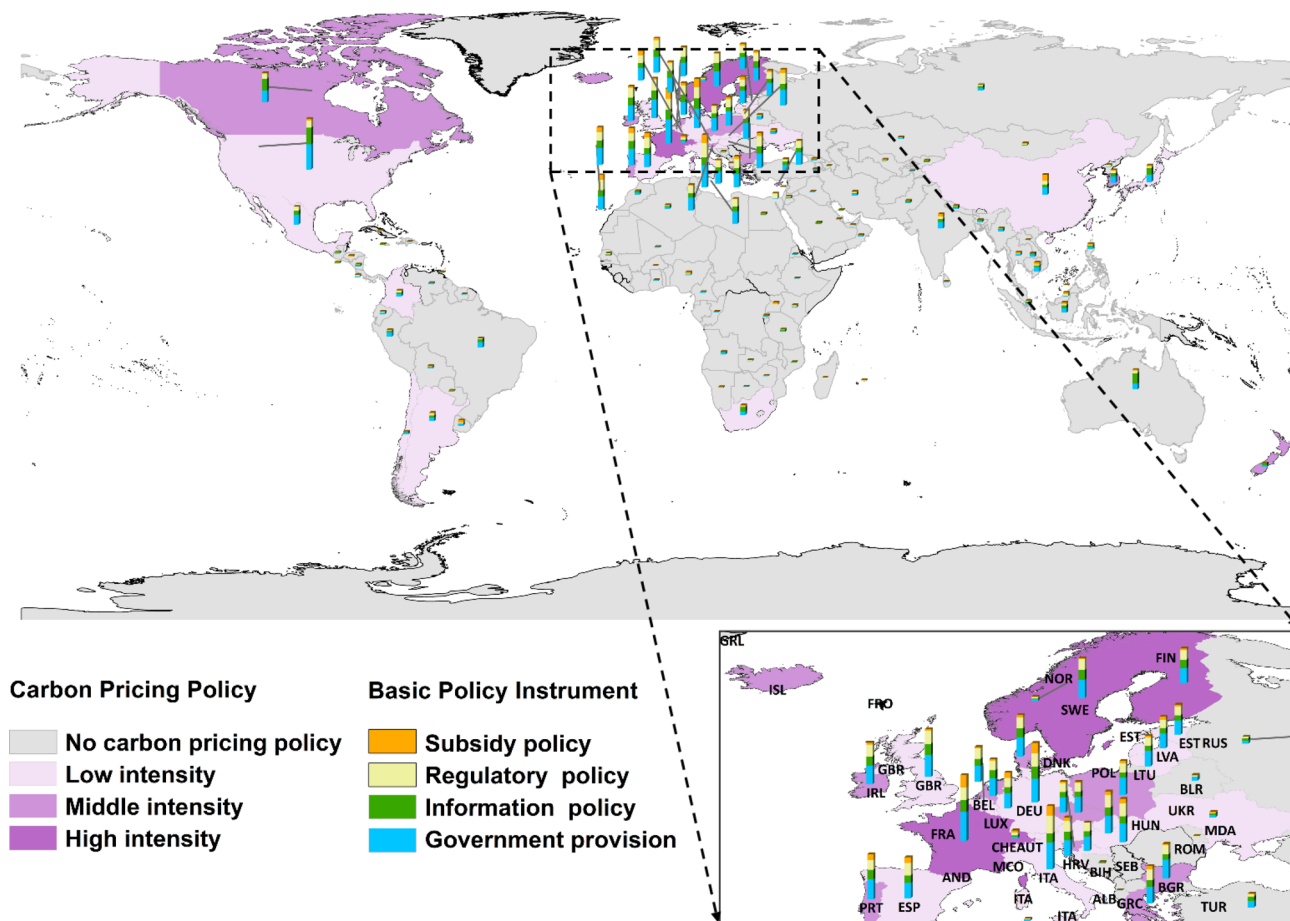
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Extended Data Fig. 1 | Analytical Framework for Assessing Synergistic and Conflicting Effects. This figure illustrates our three-step analytical framework for systematically assessing carbon pricing policy interactions. Data Sources: Global Climate Change Mitigation Policy Database (GCCMPD), World Bank

Carbon Pricing Dashboard (WBCPD), World Carbon Pricing Database (WCPD). The Robustness Check Framework ensures methodological rigor through comprehensive validation.



Extended Data Fig. 2 | National Climate Change Mitigation Policy Index for 2019. This figure reflects the global configuration of carbon pricing and complementary climate policy instruments, providing the analytical foundation for examining interaction effects between carbon pricing and complementary instruments. Superimposed stacked bars represent the relative weights of four basic policy instruments—subsidy, regulatory, information, and government provision—capturing their contributions to national mitigation portfolios.

The European inset provides a magnified view of jurisdictions with denser and more heterogeneous policy mixes, facilitating comparisons of instrument layering and interaction across countries with varying levels of carbon pricing intensity. These divergent configurations indicate that the effectiveness of carbon pricing depends not only on price levels or coverage, but also on the surrounding mix of synergistic or conflicting policies. Basemap from Natural Earth (<https://www.naturalearthdata.com>).

Extended Data Table 1 | Policy Interaction Effect on Carbon Emission Intensity

	CO ₂ Emission Intensity										
	Carbon Tax			Carbon Tax after EU ETS (4)	ETS						
	All sample (1)	Energy System & Industry (2)	Multi-sector (3)		All sample (5)	EU ETS (6)	Others				
							All (7)	Mass-based (8)	Rate-based (9)	High Intensity (10)	Low Intensity (11)
Interaction between Carbon Tax and ETS				0.002 (0.001)	-0.00003 (0.0001)	-0.00002 (0.0001)	-0.0004 (0.001)	-0.006*** (0.025)	0.054 (0.002)	0.0005 (0.002)	-0.0003 (0.005)
Interaction with other policies:											
Subsidy Policy	0.155 (0.185)	-0.073 (0.414)	-0.386 (0.740)	0.068*** (0.019)	0.027*** (0.003)	0.033*** (0.007)	0.122 (0.245)	1.026*** (0.242)	-0.314 (0.495)	0.529** (0.186)	-1.026** (0.411)
Regulatory Policy	-0.074 (0.179)	-0.268 (0.786)	1.246 (0.742)	-0.061 (0.094)	-0.039*** (0.007)	-0.038** (0.016)	-0.041 (0.054)	0.037 (0.067)	-0.084 (0.997)	-0.031 (0.022)	0.529 (0.883)
Information Policy	-0.077 (0.130)	0.065 (0.954)	0.514 (0.442)	-0.031 (0.094)	-0.001 (0.013)	-0.030* (0.017)	0.047 (0.061)	0.162*** (0.019)	-0.255 (0.343)	0.001 (0.032)	1.819*** (0.371)
Government Provision	0.133 (0.180)	0.460 (0.644)	-1.078** (0.455)	-0.033 (0.110)	0.009 (0.016)	0.016 (0.018)	-0.004 (0.149)	-0.481*** (0.126)	0.550 (0.324)	-0.118 (0.095)	-2.132*** (0.736)
Observation	33	12	21	34	485	415	70	50	20	30	40
Adjusted R-squared	0.769	0.871	0.816	0.746	0.536	0.547	0.424	0.799	0.934	0.929	0.628

This table presents the estimated coefficients and standard errors (in parentheses). The dependent variable is the log difference in CO₂ emission intensity before and after policy implementation, capturing the synthetic control–estimated effect. The data covers the period from 1996 to 2019. The interaction coefficient is reported in the “Interaction with other policies” row. In column (1) of the carbon tax results, only countries and regions that implemented carbon tax policies without an ETS are included. Denmark, Finland, Norway, Poland, Slovenia, and Sweden are excluded as they adopted carbon taxes before 1996, making them ineligible for synthetic control analysis. Columns (2) and (3) are obtained by clustering the characteristics of each carbon tax policy in (1). Column (4) includes countries that adopted ETS policies before implementing carbon taxes. The sample in column (5) includes all countries and regions that have implemented ETS policies, including those that have implemented carbon tax policies simultaneously. Column (6) focuses on the EU ETS sample. Column (7) includes non-EU ETS regions. Australia is excluded due to its short-lived carbon trading market (2012–2014), which could distort policy effect estimates. Columns (8) and (9) classify regions in (7) based on emission caps: mass-based or rate-based. Columns (10) and (11) categorize the sample in (7) by ETS policy intensity. Time-fixed effects are controlled in all regressions. * p < 0.10, ** p < 0.05, *** p < 0.01.